

Mohammed Suhail Rafeeq V Systems Software Engineer

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PROFESSIONAL EXPERIENCE

Founding Engineer, Cawosh

10/2025 – Present
UK (Remote)

- Designed and implemented backend systems, application services, and cloud infrastructure for a **multi-tenant** garage management platform.
- Led development within a small engineering team while building scalable services and deployment workflows using **Python, AWS, Docker Swarm**, and **Pulumi**.
- Designed a cost-effective containerized deployment architecture balancing operational simplicity, reliability, and infrastructure costs for an early-stage production environment.

Software Engineer, TurboML

10/2024 – 09/2025
India (Remote)

TurboML (MLOps Platform)

- Delivered a **10x Python SDK performance improvement** using **Apache Arrow**-based zero-copy ingestion integrated with **C++ execution runtimes**.
- Improved reliability and debuggability of real-time execution systems by redesigning **snapshot handling** and runtime portability mechanisms in **C++**.
- Designed runtime portability mechanisms between local in-process and remote execution runtimes for seamless deployment and debugging workflows.

Puch AI

- Helped scale the platform to support **1M+ users** and high-concurrency messaging workloads.
- Designed and implemented the initial **RAG pipeline** using **Qdrant** for low-latency retrieval and contextual response generation.
- Refactored conversation storage and retrieval workflows, improving context handling and LLM response quality.

Software Developer, SOTI

09/2020 – 12/2023
Kochi, India

- Part of the **Core Performance team** responsible for scalability improvements supporting **1M+ managed devices**.
- Designed **caching layers** for read-heavy backend services, reducing **tail latency by 3x** and improving reliability at scale.
- Contributed to **Linux device management agent development** in **C++** and built proof-of-concepts for new features, architectural changes, and performance improvements.
- Mentored engineers on **.NET CLR internals**, **C++ runtime systems**, and low-level debugging.

SKILLS

- **Database Systems** – PostgreSQL Internals, Apache Arrow, DataFusion, OLAP Systems, Query Execution, Vectorized Processing, Arrow Flight SQL.
- **Programming Languages** – Rust, C/C++, Go, Python, SQL, C#, TypeScript
- **Distributed Infrastructure & Performance** – Kubernetes, AWS, Linux, Containers, Pulumi, eBPF, Valgrind, Profiling, Performance Tuning, Debugging

PROJECTS

pgFusion

2025

- Built a high-performance OLAP engine enabling analytical queries directly on **PostgreSQL heap files** without ETL or data movement.
- Integrated **Apache DataFusion** and **Arrow Flight SQL** for vectorized query execution and analytical query offloading.
- Developed a native **Postgres-to-Arrow** reader for efficient columnar data access from PostgreSQL storage files.

NaadanDB

2024

- Built an experimental relational database exploring **storage engine design**, **transaction processing**, **page layouts**, and **query execution** for hybrid **OLTP/OLAP** workloads.
- Designed and implemented core database components, including transactional workflows, page-based storage structures, and query execution primitives.

Rusty-Fuzzer

2019

- Developed a **Rust-based** evolutionary fuzzer leveraging **LLVM Sanitizer Coverage** instrumentation for automated vulnerability discovery.

Additional Projects

- **CloudBnB**, **DesignSphere**, and other systems/infrastructure projects available on Website

EDUCATION

Bachelor of Technology, Computer Science and Engineering, School of Engineering, CUSAT. (GPA-8.4)

2016 – 2020
Kochi, India